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Shen Zhen MTC Co., LTD

TEST REPORT

SCOPE OF WORK
Radio Frequency Testing–MHDV4080-T9503-S2, MHDV4080-T9503, MHDV40****- T9503-##, MHDV40****- T9503, (*, # can from 0 to 9, A to Z or blank), 2T-C40HD2225EB, 2T-C40HD2225KB, 2T-C40HD2725KB, 2T-C40HD2725EB, 2T-C40HD3225EB, 2T-C40HD3225KB, 2T-C40HD2x25EB, 2T-C40HD2x25KB, **-*40********(* can from 0 to 9,A to Z or blank), PR40F2462VKB, F40AR5G, LT-40CR350, L40RFE25

REPORT NUMBER
260508019SZN-006

ISSUE DATE	**[REVISED DATE]**
20 May 2026	[-------------------]

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ETSI EN301893_c

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TEST REPORT

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RADIO COMMUNICATIONS TESTING REPORT

Shen Zhen MTC Co., LTD

MHDV4080-T9503-S2, MHDV4080-T9503, MHDV40****- T9503-##, MHDV40****- T9503, (*, # can from 0 to 9, A to Z or blank), 2T-C40HD2225EB, 2T-C40HD2225KB, 2T-C40HD2725KB, 2T-C40HD2725EB, 2T-C40HD3225EB, 2T-C40HD3225KB, 2T-C40HD2x25EB, 2T-C40HD2x25KB, **-*40********(* can from 0 to 9,A to Z or blank), PR40F2462VKB, F40AR5G, LT-40CR350, L40RFE25

TV/LED TV

5 GHz Transceiver

Test Report: 260508019SZN-006

Report Prepared and Checked:	Bruce Zheng Project Engineer	
Report Approved By:	Rode Liu Project Engineer	
Date:	20 May 2026	

RADIO PERFORMANCE MEASUREMENTS

RESULT SUMMARY

Test Requirement		Test Specification	Result
Centre frequencies	4.2.1	5.4.2	Pass
Nominal, and occupied, channel bandwidth	4.2.2	5.4.3	Pass
RF output power	4.2.3	5.4.4	Pass
Transmit Power Control	4.2.3	5.4.4	N/A
Power Density	4.2.3	5.4.4	Pass
Transmitter unwanted emissions outside the 5 GHz RLAN bands	4.2.4.1	5.4.5	Pass
Transmitter unwanted emissions within the 5 GHz RLAN bands	4.2.4.2	5.4.6	Pass
Receiver spurious emissions	4.2.5	5.4.7	Pass
Adaptivity	4.2.7	5.4.9	Pass
Receiver Blocking	4.2.8	5.4.10	Pass
Dynamic Frequency Selection (DFS)	4.7.6	5.4.8	N/A

*N/A: The manufacturer declared that the equipment is without TPC function.

EQUIPMENT UNDER TEST (EUT)

INFORMATION

Applicant:	Shen Zhen MTC Co., LTD MTC Industry Park, 1st Lilang Road, Xialilang community, Nanwan Longgang district, Shenzhen, China
Description of EUT:	TV/LED TV
Type Number (s):	MHDV4080-T9503-S2
Brand Name(s):	AMTC, SHARP, Polariod, VITADO, JVC, Logik
Serial Number (s):	Not Labelled
Equipment Received:	14 January 2025
Test Date (s):	14 January 2025 to 12 February 2025
Operating Mode(s):	5150~5250 MHz
Test Modulation:	802.11a: OFDM 802.11n:OFDM
Test Site and Location:	Intertek Testing Services Shenzhen Ltd. No.101&201, Building B, No.308, Wuhe Avenue, Zhangkengjing, Guanhu Subdistrict, Longhua District, Shenzhen, Guangdong, China
Test Specification (s):	ETSI EN 301 893 V2.1.1 (2017-05)

CONTENTS

EXHIBIT 1

GENERAL DESCRIPTION

1 INTRODUCTION

Intertek Testing Services Shenzhen Limited has tested the Shen Zhen MTC Co., LTD MHDV4080-T9503-S2, TV/LED TV. The sample was tested to the relevant performance specification published by the European Telecommunications Standards Institute. This report contains the results of these tests and is submitted to Shen Zhen MTC Co., LTD as the final test results.

The Model: MHDV4080-T9503, MHDV40****- T9503-##, MHDV40****- T9503, (*, # can from 0 to 9, A to Z or blank), 2T-C40HD2225EB, 2T-C40HD2225KB, 2T-C40HD2725KB, 2T-C40HD2725EB, 2T-C40HD3225EB, 2T-C40HD3225KB, 2T-C40HD2x25EB, 2T-C40HD2x25KB, **-*40********(* can from 0 to 9,A to Z or blank), PR40F2462VKB, F40AR5G, LT-40CR350, L40RFE25 are the same as the Model: MHDV4080-T9503-S2 in hardware aspect. The difference in model number serves as marketing strategy.

This report bases on the previous report with report number 251219032SZN-004 Dated 09 January 2026 (original signature by Bruce Zheng, Rode Liu on file). Owing to adding the models, no test was required after evaluation.

Due to the use of the same wireless module, Intertek only conducted the Radiated Spurious Emissions test. The results of other test items were cited from the module report CN23DLMM 002 Date of issue is 27 October 2023 (original signature by Breeze Jiang, Lin Lin on file).

The production units are required to conform to the initial sample as received when the units are placed on the market.

2 TEST SPECIFICATION

2.1 RELEVANT PERFORMANCE SPECIFICATION

The relevant performance specifications for Shen Zhen MTC Co., LTD MHDV4080-T9503-S2 TV/LED TV are ETSI EN 301 893 V2.1.1 (2017-05). The harmonised standards are ETSI EN 301 893 V2.1.1 (2017-05).
The tests performed are those required to demonstrate compliance with the technical specifications and the essential requirements of Article 3.2 of the Radio Equipment Directive (2014/53/EU)-RED for regulatory purposes.

2.2 TEST ENVIRONMENT

The tests were performed in the Radio communications and Electromagnetic Compatibility Test Facility at Intertek Testing Services Shenzhen Limited. The sample was subjected to the ambient conditions in the laboratory and indoor test site except during tests at extremes of temperatures and the Radiated Emissions Tests. The temperature and relative humidity recorded during the period of each test are given in the results.

2.3 CONFIGURATION OF TEST SAMPLE

The test sample consisted of one transceiver and under manufacturer's specified test mode, only the worst case data were reported.

2.4 TEST POWER SOURCES

The TV/LED TV is intended to operate from 220-240V~, 50/60Hz, 70W or 90W. The test power source voltages were:

Nominal test voltage AC 230V

2.5 TEST FREQUENCIES

U-NII-1:

Band 5150~5250MHz			
802.11a / n(HT20)		802.11n(HT40)	
Channel	Frequency (MHz)	Channel	Frequency (MHz)
36	5180	38	5190
40	5200	46	5230
44	5220		
48	5240		

2.6 APPLIED TEST MODES AND CHANNELS

Test	Clause	Test channels		
		Lower sub-band (5150 MHz to 5350 MHz)		Higher sub-band 5470 MHz to 5725 MHz
		5150 MHz to 5250 MHz	5250 MHz to 5350 MHz	
Centre frequencies	5.4.2	C7 (see note 1)		C8 (see note 1)
Occupied Channel Bandwidth	5.4.3	C7		C8
Power, power density	5.4.4	C1	C2	C3, C4
Transmitter unwanted emissions outside the 5 GHz RLAN bands	5.4.5	C7 (see note 1)		C8 (see note 1)
Transmitter unwanted emissions within the 5 GHz RLAN bands	5.4.6	C1	C2	C3, C4
Receiver spurious emissions	5.4.7	C7 (see note 1)		C8 (see note 1)
Transmit Power Control (TPC)	5.4.4	n.a. (see note 2)	C2 (see note 1)	C3, C4 (see note 1)
Dynamic Frequency Selection (DFS)	5.4.8	n.a. (see note 2)	C5	C6 (see note 3)
Adaptivity	5.4.9	C9		
Receiver Blocking	5.4.10	C7		C8

C1, C3: The lowest declared channel for every declared nominal channel bandwidth within this band.

C2, C4: The highest declared channel for every declared nominal channel bandwidth within this band.

C5, C6: One channel out of the declared channels for this frequency range. If more than one nominal channel bandwidth has been declared for this sub-band, testing shall be performed using the lowest and highest nominal channel bandwidth.

C7, C8: One channel out of the declared channels for this sub-band. For Occupied Channel Bandwidth, testing shall be repeated for every declared nominal channel bandwidth within this sub-band. For Adaptivity, testing shall be performed using the highest nominal channel bandwidth.

C9: One channel (in case of single-channel testing) or a group of channels (in case of multi-channel testing) out of the declared channels.

Note :

(1) In case of more than 1 channel plan has been declared, testing of these specific requirements need only be performed using one of the declared channel plans.

(2) Testing is not required for nominal channel bandwidths that fall completely within the frequency range 5150 MHz to 5250 MHz.

(3) Where the declared channel plan includes channels whose nominal channel bandwidth falls completely or partly within the 5600 MHz to 5650 MHz band, the tests for the *Channel* Availability Check (and where implemented, for the Off-Channel CAC) shall be performed on one of these channels in addition to a channel within the band 5470 MHz to 5600 MHz or 5650 MHz to 5725 MHz band.

2.7 GENERAL REQUIREMENTS

2.7.1 Modulation

Constant envelope digital frequency modulation is used with OFDM.

2.7.2 Antenna

The antenna used in Transceiver is Integral antenna.

Antenna Gain: 2.57dBi.

2.7.3 Test Condition

Normal test condition: NTNV (NT: Normal Temperature; NV: Normal voltage).

Extreme test condition: according to clause 5.1.3, extreme condition refer to the extreme temperature.

Environment Parameter	Selected Values During Tests	
Relative humidity content:	55 %	
Temperature:	LT	-10 °C
	NT	25 °C
	HT	55 °C
Power supply:	NV	AC 230/50Hz
Air pressure:	not relevant for this kind of testing	
Note: 1. TL: Low extreme Temperature. 2. TH: High extreme Temperature. 3. Temperature range is declared by the manufacturer.		

2.8 MEASUREMENT UNCERTAINTY

All measurement uncertainties stated in this report are estimated to a 95% confidence level.

2.9 SUPPORT EQUIPMENT – RADIO PERFORMANCE MEASUREMENTS

1. Laptop (Dell 5420) (Provided by Intertek)
2. USB Cable (Detachable, un-shielded with core, 100cm) (Provided by Applicant)
3. Test Software (Provided by Applicant) was installed on the EUT

EXHIBIT 2

TEST RESULT
OF
RADIO PERFORMANCE MEASUREMENTS

3 TRANSMITTER UNWANTED EMISSIONS OUTSIDE THE 5GHZ RLAN BANDS

3.1 Test Method and Summary

Basic Standard:	ETSI EN 301 893 V2.1.1 (2017-05)
Clause:	5.4.5
Test method	Radiated measurements

3.2 Equipment List

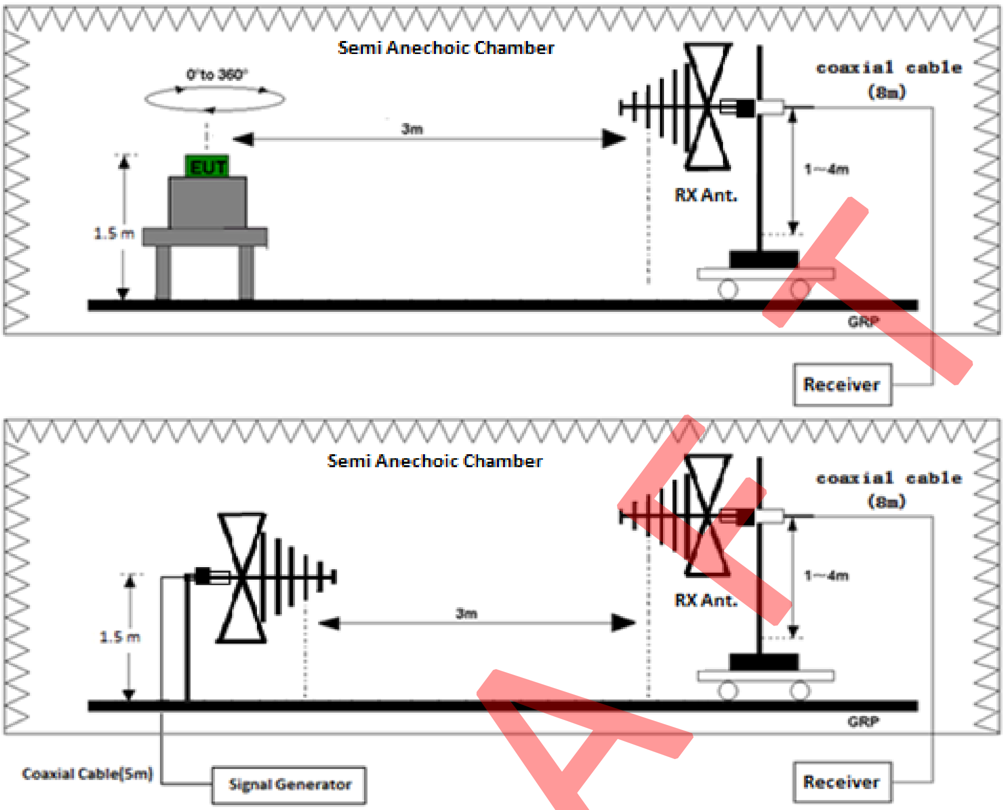
Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ185-03	EMI Receiver	R&S	ESR7	2024-04-23	2025-04-23
SZ056-06	Signal Analyzer	R&S	FSV40	2024-12-06	2025-12-06
SZ061-13	Biconilog Antenna	ETS	3142E	2022-07-13	2025-07-13
SZ061-08	Horn Antenna	ETS	3115	2024-09-13	2027-09-13
SZ181-08	Microwave System Amplifier	keysight	83017A	2024-07-29	2025-07-29
SZ188-05	Anechoic Chamber	ETS	FACT 3-2.0	2021-05-25	2026-05-25
SZ062-40	RF Cable	Talent Microwave	A50-3.5M3.5M-4.5M	2024-09-30	2025-09-30
SZ062-41	RF Cable	Talent Microwave	A50-3.5M3.5M-8M	2024-09-30	2025-09-30
SZ062-34	RF Cable	Talent Microwave	A50-3.5M3.5M-1M	2024-09-30	2025-09-30

3.3 Limit

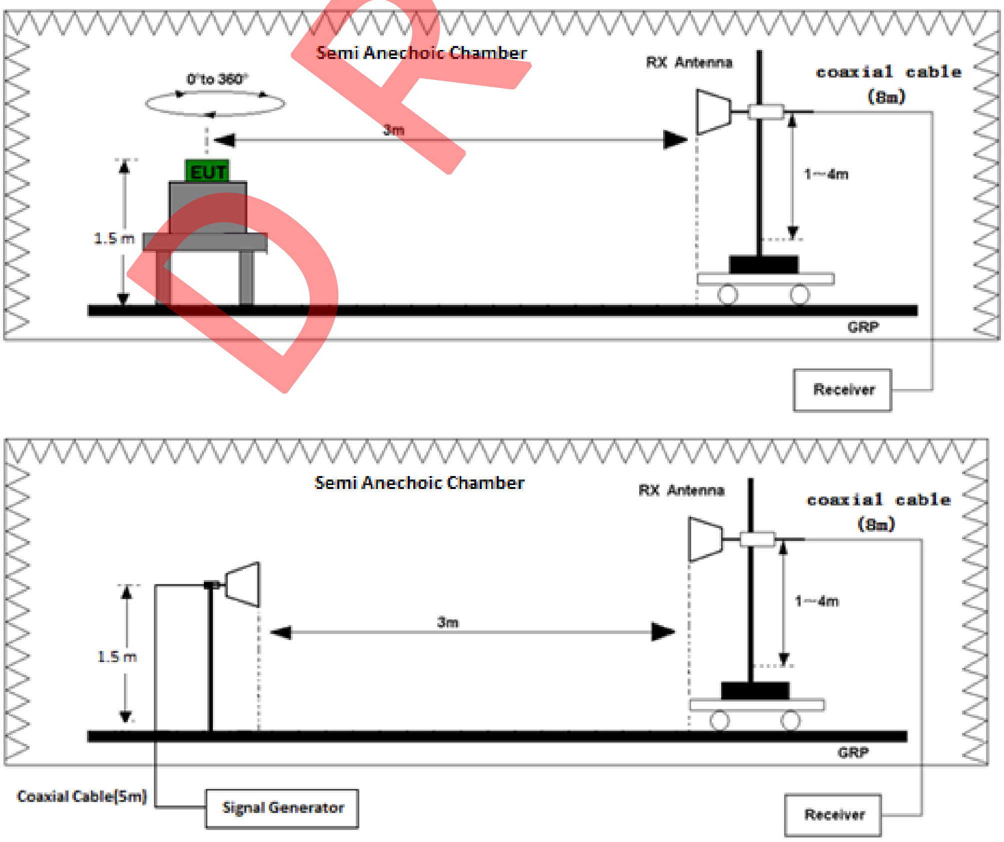
The transmitter unwanted emissions in the spurious domain shall not exceed the values.

Frequency Range	Maximum power	Bandwidth
30 MHz to 47 MHz	-36 dBm	100 kHz
47 MHz to 74 MHz	-54 dBm	100 kHz
74 MHz to 87,5 MHz	-36 dBm	100 kHz
87,5 MHz to 118 MHz	-54 dBm	100 kHz
118 MHz to 174 MHz	-36 dBm	100 kHz
174 MHz to 230 MHz	-54 dBm	100 kHz
230 MHz to 470 MHz	-36 dBm	100 kHz
470 MHz to 862 MHz	-54 dBm	100 kHz
862 MHz to 1 GHz	-36 dBm	100 kHz
1 GHz to 5.15 GHz	-30 dBm	1 MHz
5,35 GHz to 5,47 GHz	-30 dBm	1 MHz
5,725 GHz to 26 GHz	-30 dBm	1 MHz

3.4 Test Setup



Test set-up of radiated disturbance (30MHz-1GHz)



Test set-up of radiated disturbance (above 1GHz)

3.5 Measuring Instrument Setting

Below 1000MHz Pre-Scan

Spectrum Analyzer	Setting
Attenuation	Auto
Start Frequency	30 MHz
Stop Frequency	1 GHz
RBW / VBW	100KHz / 300KHz
Detector	Positive Peak
Sweep Time	For non continuous transmissions (duty cycle less than 100 %), the sweep time shall be sufficiently long, such that for each 100 kHz frequency step, the measurement time is greater than two transmissions of the UUT.
Sweep Point	≥9700

Above 1000MHz Pre-Scan

Spectrum Analyzer	Setting
Attenuation	Auto
Start Frequency	1 GHz
Stop Frequency	26 GHz
RBW / VBW	1MHz / 3MHz
Detector	Positive Peak
Sweep Time	For non-continuous transmissions (duty cycle less than 100 %), the sweep time shall be sufficiently long, such that for each 1 MHz frequency step, the measurement time is greater than two transmissions of the UUT.
Sweep Point	≥25000

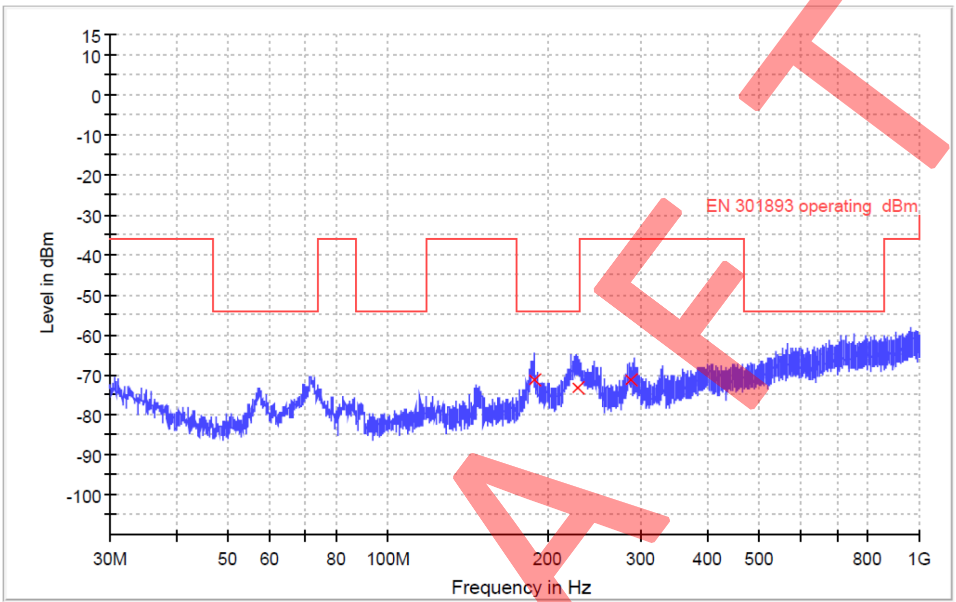
The Spurious emissions identified during the Pre-Scan

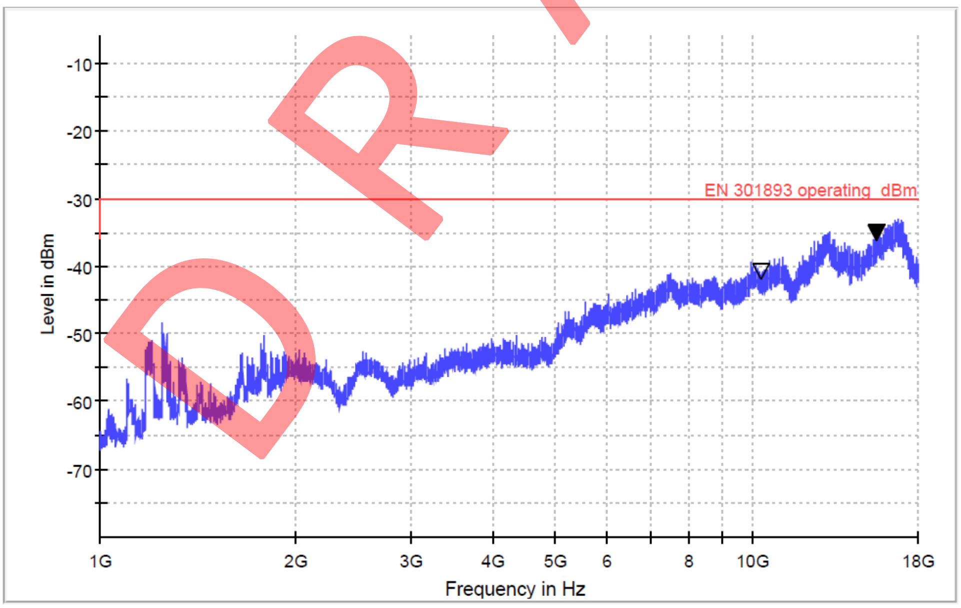
Spectrum Analyzer	Setting
Attenuation	Auto
RBW	100KHz (< 1GHz), 1MHz (>1GHz)
VBW	300KHz (< 1GHz), 3MHz (>1GHz)
Detector	RMS
Span	0
Sweep Time	Suitable to capture one transmission burst. Additional measurements may be needed to identify the length of the transmission burst. In case of continuous signals, the Sweep Time shall be set to 30 ms
Sweep Point	Sweeptime [µs] / 1 µs with a maximum of 30 000

3.6 Worse Case Test Result (AC Power Supply)

Radiated Measurement

802.11a -Worse case
5180MHz
Antenna Polarization Horizontal:





Limit and Margin

Frequency (MHz)	RMS (dBm)	Meas. Time (ms)	Pol	Margin - RMS (dB)	Limit - RMS (dBm)
188.886000	-71.3	1000.0	H	17.3	-54.0
10360.000000	-41.7	1000.0	H	11.7	-30.0
15540.000000	-36.0	1000.0	H	6.0	-30.0

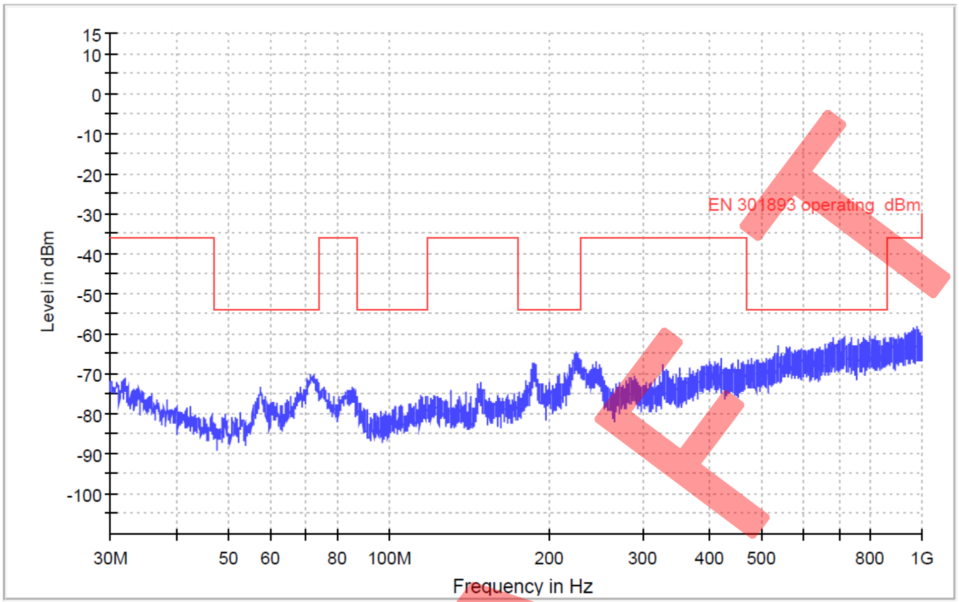
☐ **No emissions significantly above equipment noise floor.**

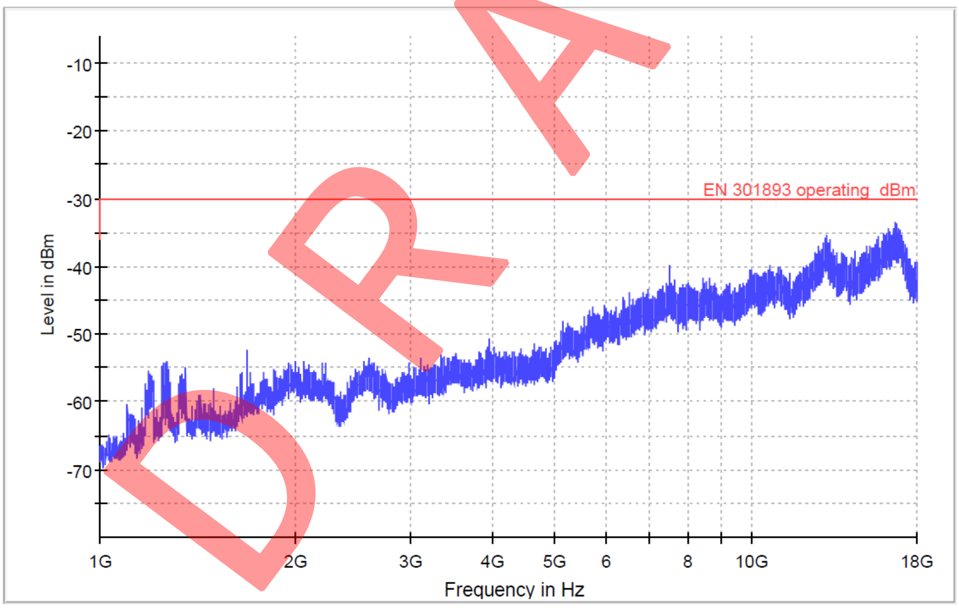
Notes:

1. The test frequency range is 30MHz to 26GHz.
2. Other emissions found were at least 10 dB below the limit.
3. Measurement Uncertainty: ±4.8dB for frequency range 30MHz to 1GHz and ±6.0dB for frequency range 1GHz to 26GHz.

5240MHz

Antenna Polarization Horizontal:





Limit and Margin

Frequency (MHz)	RMS (dBm)	Meas. Time (ms)	Pol	Margin - RMS (dB)	Limit - RMS (dBm)
227.391000	-69.5	1000.0	H	15.5	-54.0
10480.000000	-42.3	1000.0	H	12.3	-30.0
15720.000000	-38.0	1000.0	H	8.0	-30.0

☐ **No emissions significantly above equipment noise floor.**

Notes:

1. The test frequency range is 30MHz to 26GHz.
2. Other emissions found were at least 10 dB below the limit.
3. Measurement Uncertainty: ±4.8dB for frequency range 30MHz to 1GHz and ±6.0dB for frequency range 1GHz to 26GHz.

4 RECEIVER SPURIOUS EMISSIONS

4.1 Test Method and Summary

Basic Standard:	ETSI EN 301 893 V2.1.1 (2017-05)
Clause:	5.4.7
Test method	Radiated measurements and Conducted measurements
Test Mode	Receive-only mode

4.2 Equipment List

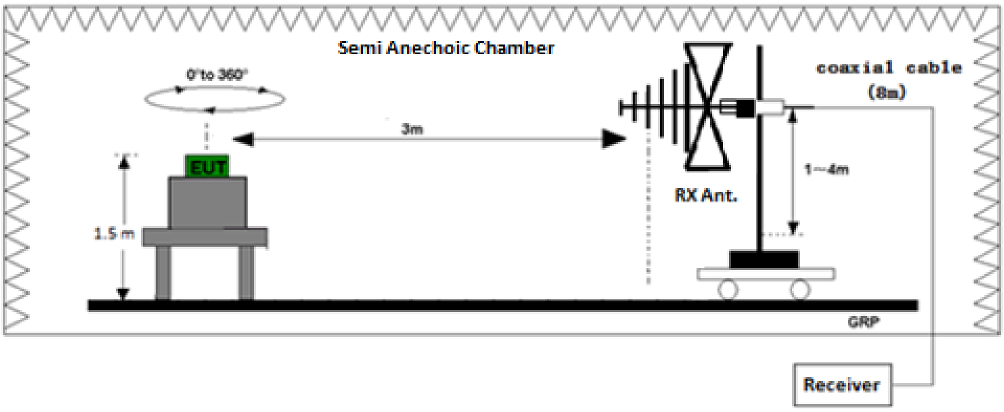
Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ185-03	EMI Receiver	R&S	ESR7	2024-04-23	2025-04-23
SZ056-06	Signal Analyzer	R&S	FSV40	2024-12-06	2025-12-06
SZ061-13	Biconilog Antenna	ETS	3142E	2022-07-13	2025-07-13
SZ061-08	Horn Antenna	ETS	3115	2024-09-13	2027-09-13
SZ181-08	Microwave System Amplifier	keysight	83017A	2024-07-29	2025-07-29
SZ188-05	Anechoic Chamber	ETS	FACT 3-2.0	2021-05-25	2026-05-25
SZ062-40	RF Cable	Talent Microwave	A50-3.5M3.5M-4.5M	2024-09-30	2025-09-30
SZ062-41	RF Cable	Talent Microwave	A50-3.5M3.5M-8M	2024-09-30	2025-09-30
SZ062-34	RF Cable	Talent Microwave	A50-3.5M3.5M-1M	2024-09-30	2025-09-30

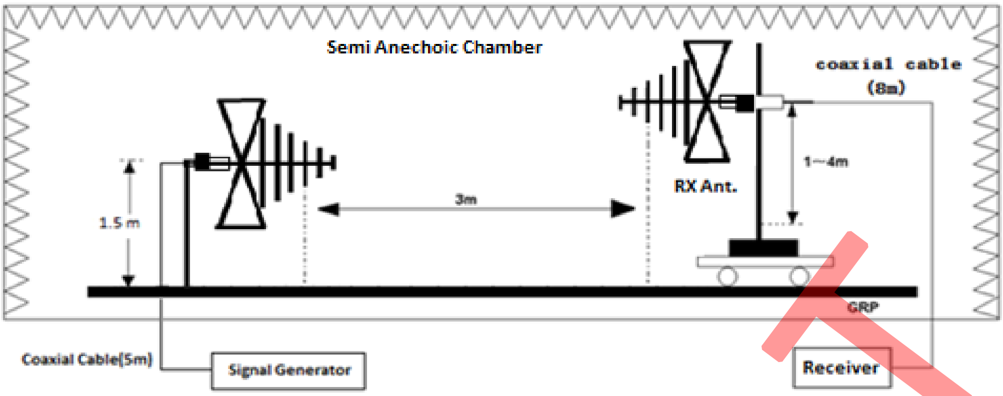
4.3 Limit for Receiver spurious emissions

The transmitter unwanted emissions in the spurious domain shall not exceed the values.

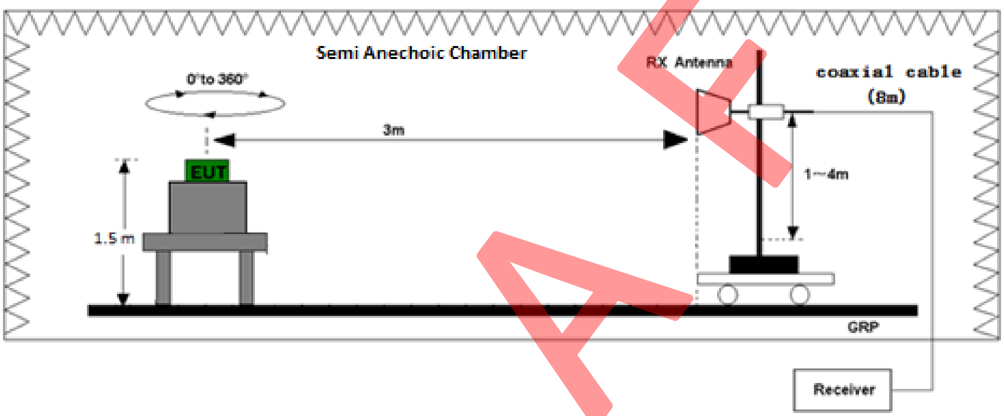
Frequency Range	Maximum power	Bandwidth
30 MHz to 1 GHz	-57 dBm	100 kHz
1 GHz to 26 GHz	-47 dBm	1MHz

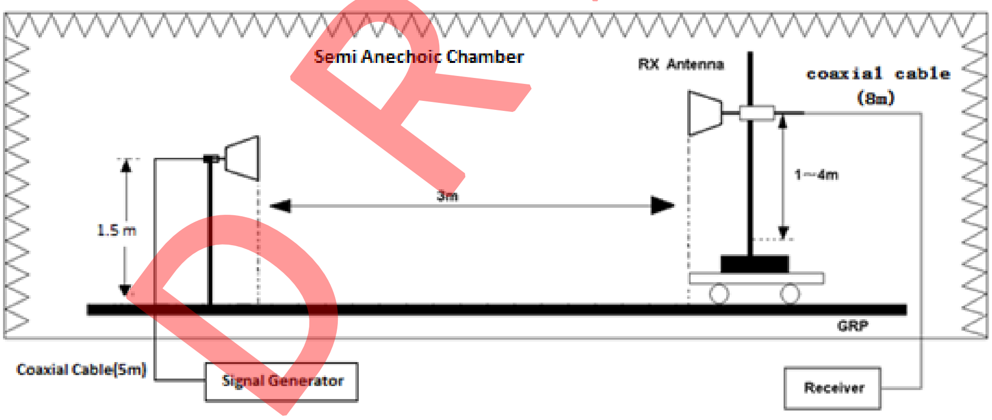
4.4 Test Setup





Test set-up of radiated disturbance (30MHz-1GHz)





Test set-up of radiated disturbance (above 1GHz)

4.5 Measuring Instrument Setting

Below 1000MHz Pre-Scan

Spectrum Analyzer	Setting
Attenuation	Auto
Start Frequency	30 MHz
Stop Frequency	1 GHz
RBW / VBW	100KHz / 300KHz
Detector	Positive Peak
Sweep Time	Auto
Sweep Point	≥9700

Above 1000MHz Pre-Scan

Spectrum Analyzer	Setting
Attenuation	Auto
Start Frequency	1 GHz
Stop Frequency	26 GHz
RBW / VBW	1MHz / 3MHz
Detector	Positive Peak
Sweep Time	Auto
Sweep Point	≥25000

Identified during the Pre-Scan

Spectrum Analyzer	Setting
Attenuation	Auto
RBW	100KHz (< 1GHz), 1MHz (>1GHz)
VBW	300KHz (< 1GHz), 3MHz (>1GHz)
Detector	RMS
Span	0
Sweep Time	30ms
Sweep Point	≥30000

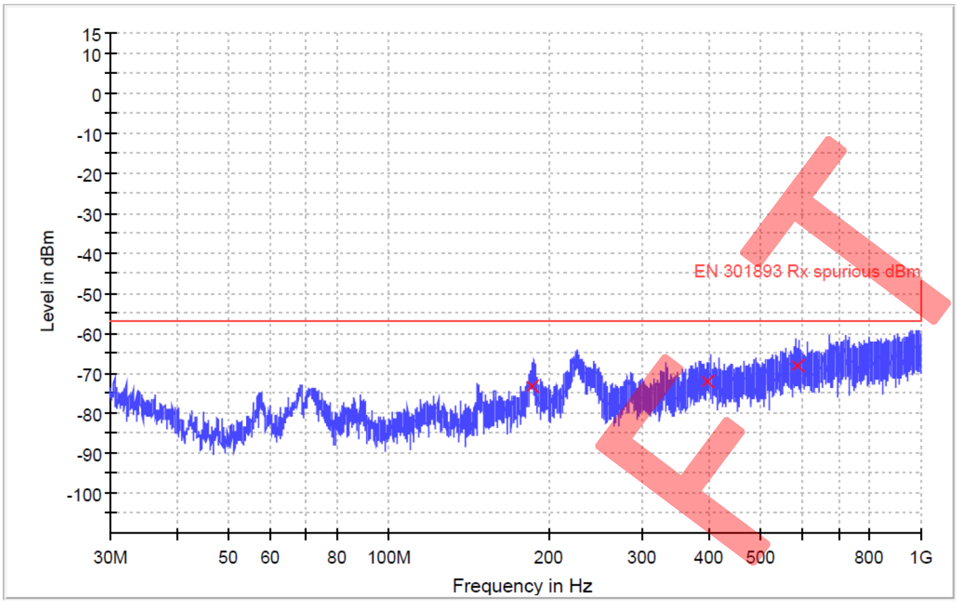
4.6 Test Results

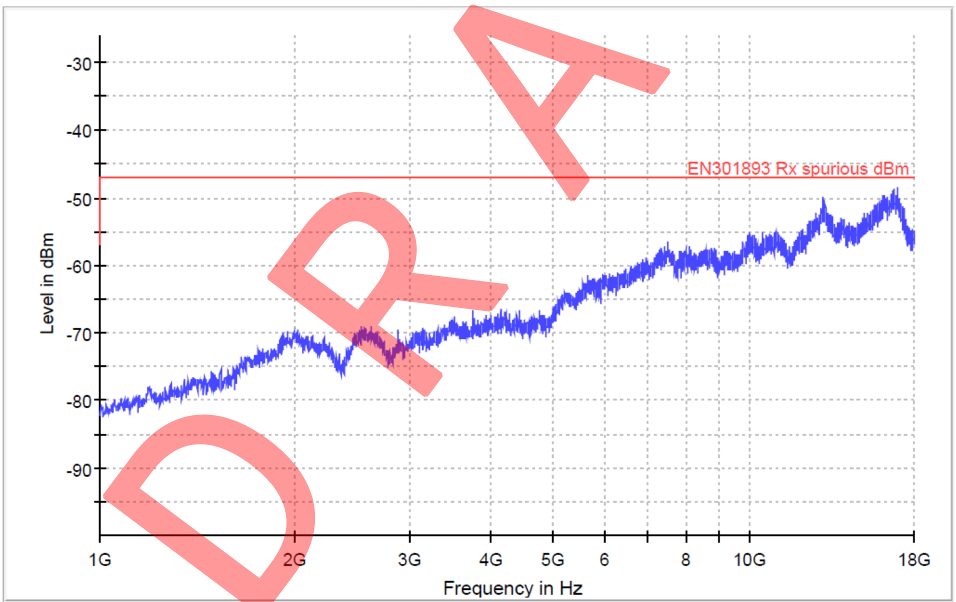
Radiated Measurement

Spurious Emissions – Standby
There were no emissions found above system measuring level (at least 10 dB below the limit).

Receiver Spurious Emissions – Operating

802.11a **-Worse case-ANT1**
5180MHz
Antenna Polarization Horizontal:





Limit and Margin

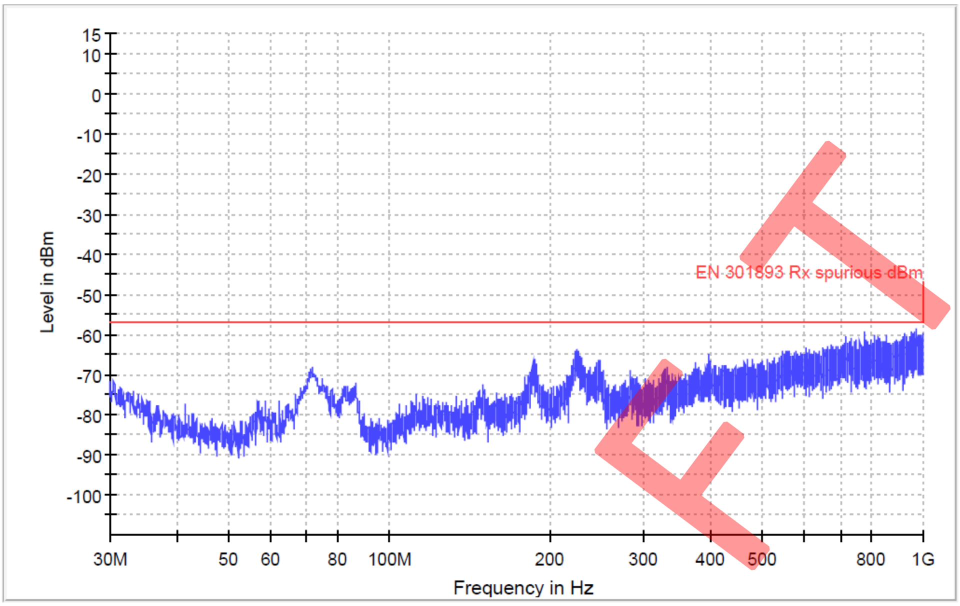
Frequency (MHz)	RMS (dBm)	Meas. Time (ms)	Pol	Margin - RMS (dB)	Limit - RMS (dBm)
586.780000	-68.1	1000.0	H	11.1	-57.0
10480.000000	-61.3	1000.0	H	14.3	-47.0
16283.000000	-53.1	1000.0	H	6.1	-47.0

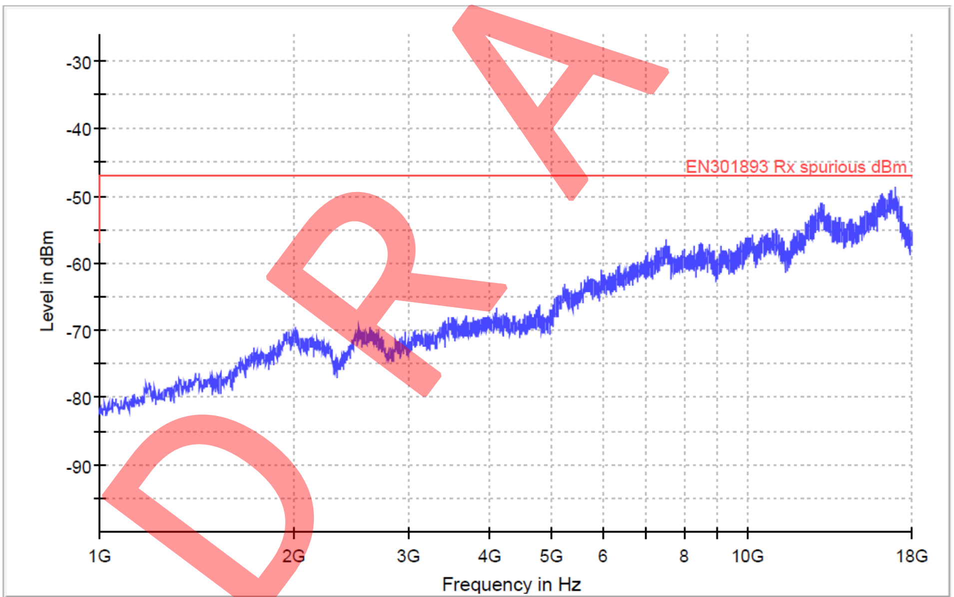
☐ No emissions significantly above equipment noise floor.

Notes:

1. The test frequency range is 30MHz to 26GHz.
2. Other emissions found were at least 10 dB below the limit.
3. Measurement Uncertainty: ±4.8dB for frequency range 30MHz to 1GHz and ±6.0dB for frequency range 1GHz to 26GHz.

5240MHz
Antenna Polarization Horizontal:





Limit and Margin

Frequency (MHz)	RMS (dBm)	Meas. Time (ms)	Pol	Margin - RMS (dB)	Limit - RMS (dBm)
238.690000	-69.1	1000.0	H	12.1	-57.0
13260.000000	-59.3	1000.0	H	12.3	-47.0
16396.000000	-54.3	1000.0	H	7.3	-47.0

☐ No emissions significantly above equipment noise floor.

Notes:

1. The test frequency range is 30MHz to 26GHz.
2. Other emissions found were at least 10 dB below the limit.
3. Measurement Uncertainty: ±4.8dB for frequency range 30MHz to 1GHz and ±6.0dB for frequency range 1GHz to 26GHz.

EXHIBIT 3

PHOTOS OF EUT

5 PHOTOS OF TEST SET-UP

Please refer to Annex of EUT Photos_260508019SZN.

*********************************** End of Report************************************